The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR A single **linear map** on residual activations predicts an LLM's answer from its context **before generation** held-out R^2 0.75, and rank-1 retrieval of the true answer among 9,941 candidates at 0.98, at the fresh-draw ceiling is already ∼87% present in the **base model**, is **shared between the assistant and fictional characters**, and predicts **sycophancy, hallucination, and misalignment before inference**.

MOTIVATION

- LLMs are trained as **next-token predictors**, but the weights fix a deterministic mapping from a context to a distribution over *whole answers*: they are also **next-answer predictors**. Absent tool results, **everything about the answer distribution is fixed before the first token is generated.**
- A **simple approximation** of that mapping would be extremely useful: predict behavior *before* generation, and predict how fine-tuning will change it.
- But it need not exist: the mapping is billions of parameters applying dozens of nonlinear layers. Prior work sits at two ends of a **capacity–usefulness tradeoff** cheap but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a whole-distribution imitator with no capacity advantage over the model itself.
- **Our question:** can a **very low-capacity** approximation capture the mapping *as a whole*? Such an approximation is a **metamodel**: a much smaller model that reads the LLM's internals and predicts its behavior.

THE RESIDUAL CONTEXT-ANSWER MAPPING

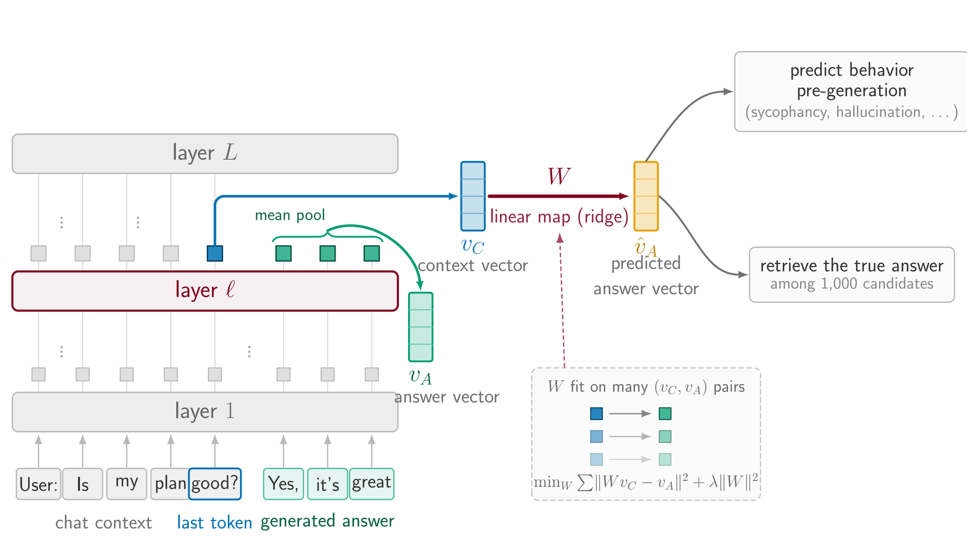

Figure 1. The **context vector** v_C is the residual-stream state at the last context token (layer ℓ); the **answer vector** v_A is the mean answer-token state. Both summaries follow prior work. *Fitted maps* $v_C \mapsto v_A$ approximate the mapping; readouts consume $\hat{v}_A$.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

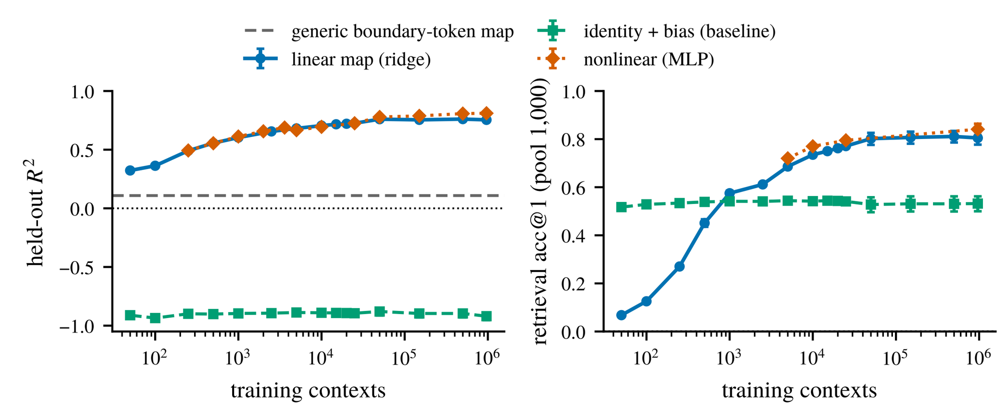

Figure 2. Held-out R^2 (left) and rank-1 retrieval (right, *raw euclidean*, pool 1,000) against training-set size, for the linear map, a nonlinear (MLP) map, and the identity + bias baseline. Dashed: a generic boundary-token control map. Qwen2.5-7B-Instruct, layer 19. Raw euclidean understates retrieval: on the multi-turn map, a whitened-cosine + CSLS read lifts rank-1 from 0.82 to 0.98 (pool 9,941).

- **High predictive power:** held-out R^2 0.75 at 963k training contexts, and rank-1 retrieval far above chance (10^{-3}) at every training size.
- **Mostly linear:** the nonlinear (MLP) map adds only ∼0.06 R^2 over the linear map, and only at large n.
- **Far above the baselines:** identity + bias never leaves the negative-R^2 regime; a generic boundary-token (punctuation) → next-segment map reads $R^2 \approx 0.11$ (dashed) so this is not just residual-stream smoothness.
- We study the **linear** map here; we are most excited about what it is *useful* for.

2. IT PREDICTS THE PERSONA-VECTOR DIRECTIONS

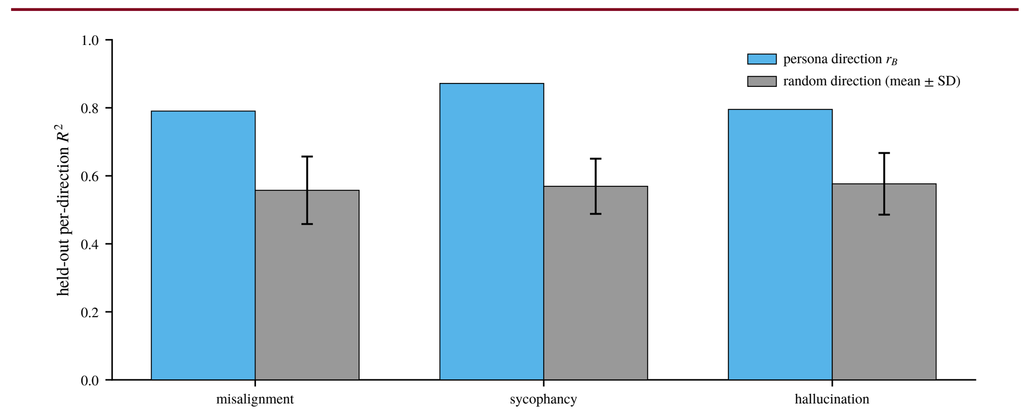

Figure 3. Held-out R^2 of the fitted map along the persona direction r_B for each behavior, against the mean ± 1 SD over 50 *random* unit directions (a spread, not a CI). Fold 0 of 5 ($n_{\text{train}} = 4{,}000$, $n_{\text{test}} = 1{,}000$); per-behavior layers 14 / 26 / 17.

- **Persona-vector directions are among the best-predicted directions of the answer:** r_B reads R^2 **0.79 / 0.87 / 0.80** (misalignment / sycophancy / hallucination) against a random-direction band of **0.56–0.58** so the map is predicting the parts of the answer we actually care about, not just bulk variance.

3. WHAT MAKES A FEATURE PREDICTABLE?

We map the context to the answer's **SAE features**, then partial out predictors *one at a time* to see which survives.

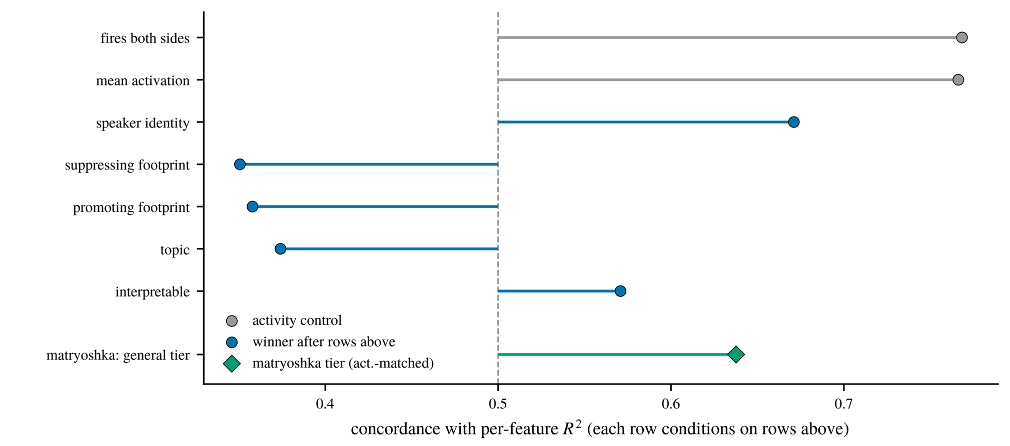

Figure 4. Forward-stepwise partialling: each row is that round's winning predictor, conditioned on *every row above it* (coarsened exact matching); c is concordance on the AUROC scale, 0.5 = chance. Gray rows are the two activity controls removed first setup, not findings. Diamond: the matryoshka *general* tier, a different dictionary and a single-control read direction only, and not part of the selection. No CIs exist here; gaps under ∼0.02 are unreadable and the conditioning coarsens down the panel. Rounds 7–13 omitted (all within ±0.07 of chance).

- **Persona survives the partialling:** once both activity controls are removed, **speaker identity** is the strongest surviving predictor (c 0.67), and the matryoshka *general* tier points the same way (0.64).
- **Low-level and content features are predicted *worse* than chance:** logit-footprint (0.35, 0.36) and topic (0.37) all fall below 0.5.
- **Concretely:** best-predicted are abstract nouns and nominalized concepts (R^2 0.95); worst are 2–4-character word fragments and single-letter detectors ($R^2 < 0$).

4. WHAT DOES IT FAIL TO RETRIEVE?

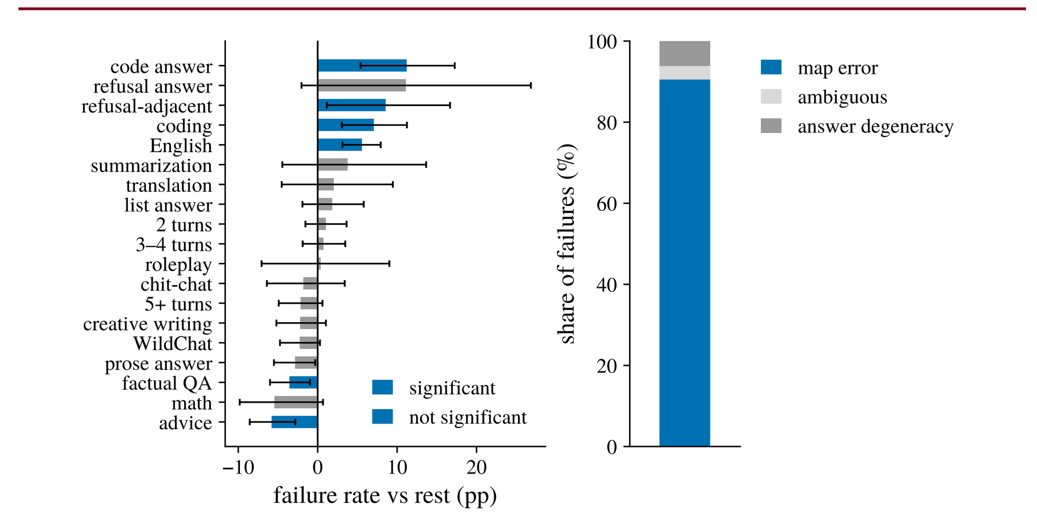

Figure 5. Retrieval failures against *sampling-noise-free* answer targets (each answer averaged over 5 on-policy draws), raw-euclidean read: $n = 1{,}988$ held-out contexts, pool 9,941, 180 failures, rank-1 0.909. Left: category failure rate minus the rest, 10,000-draw bootstrap CIs; *significant* = clears Benjamini–Hochberg FDR at $q = 0.05$. "Refusal answer" trends high but misses FDR ($n = 30$). Right: what the 180 misses are *map error* = the answer is still retrievable from a fresh draw of itself.

- **Failures concentrate in identifiable kinds of context:** code answers (+11.2pp), refusal-adjacent requests (+8.5), coding (+7.0) and English (+5.6) fail most; factual QA (−3.5) and advice (−5.8) fail least.

5. WHERE DOES THE MAPPING COME FROM?

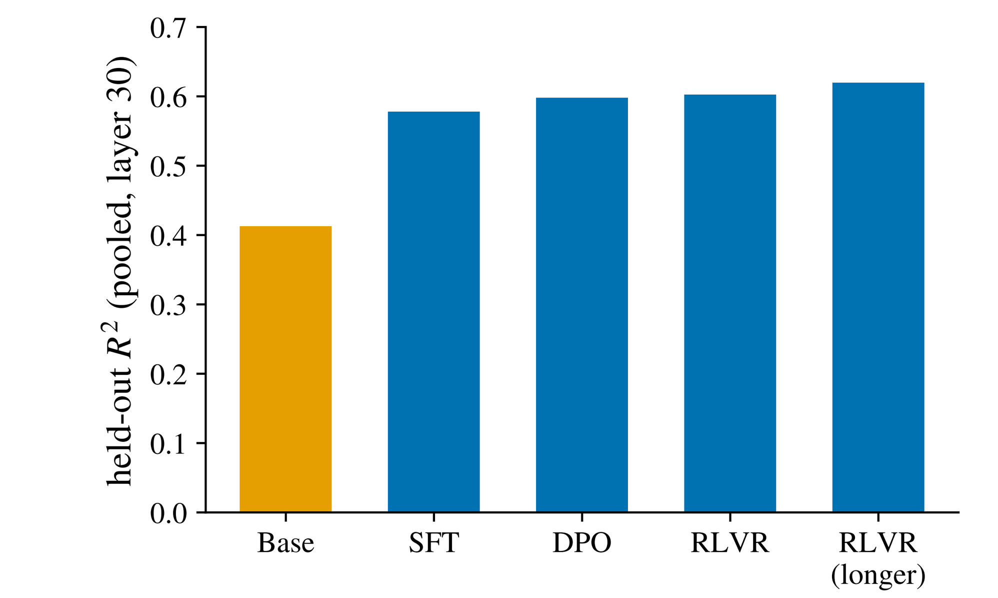

Figure 6. Pooled held-out R^2 of the fitted map at each stage of the open Llama/Tülu post-training ladder (layer 30), each stage fit and evaluated on its *own* on-policy answers.

- **Already largely present in the base model:** the base bar reads R^2 0.41 before any post-training.
- **SFT is by far the largest step** (+0.17); everything after it is essentially flat post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

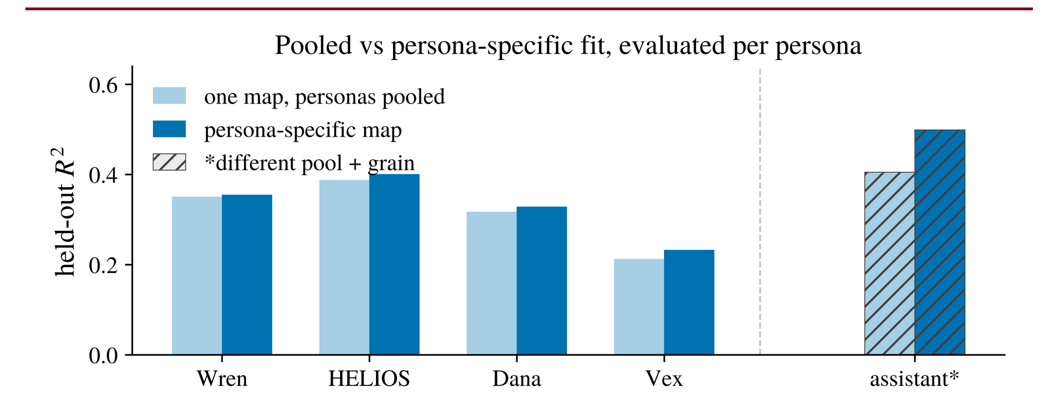

Figure 7. Held-out R^2 per character: *one* ridge map fitted on all four characters pooled (1,200 points) against that character's *own* 300-point map. Scenario-grouped 5-fold, layer 19, Qwen-2.5-7B-Instruct. **assistant*** comes from a separate committed lattice pool = {assistant Q&A, Wren Q&A, Wren in scene}, per-question rows ($n = 4{,}375$) and is shown *for reference only*: levels and fractions do not transfer between the two lattices (Wren, present in both, moves 0.99 to 0.82 on the switch alone).

- **Pooling across characters costs almost nothing:** a single map fitted on all four captures **90–98%** of what each character's own map achieves four characters, close to one shared operator.
- **The assistant behaves the same way inside its own lattice:** one pooled map captures ∼**81%** of its own-map ceiling, against 82% and 86% for the Wren cells sharing that pool unremarkable, not privileged.
- **Close, but not identical:** the character-specific edge is small and survives pooling *approaches* rather than equals a per-character fit.

APPLICATION: BEHAVIOR PREDICTION BEFORE INFERENCE

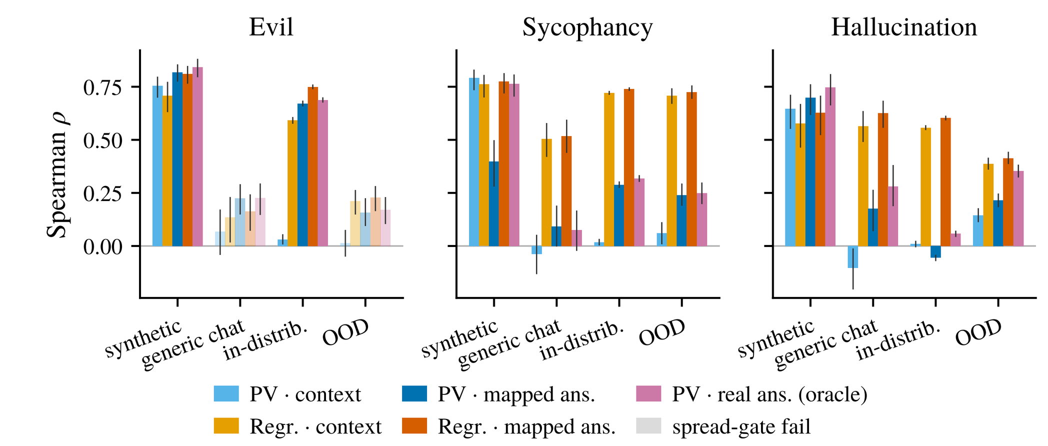

Figure 8. Spearman ρ between a readout and the judged behavior, per regime. Five readouts: the persona vector on the *context*; a fitted *regression* on the context; the persona vector on the *map-predicted* answer; a regression on the map-predicted answer; and the persona vector on the *real* answer (oracle it needs the answer generated). Regression arms consume the labelled budget; persona-vector arms are label-free. Muted bars fail the spread gate and are not interpretable. Hallucination in-distribution and OOD score fabricated fraction a different construct.

- **The best pre-inference readout is one on the *mapped answer* in every trustworthy generic-chat and OOD cell,** beating the best context-side readout by +0.01 to +0.07 though the two sycophancy margins sit inside overlapping CIs.
- **Regression on the mapped answer matches or beats regression on the context everywhere,** and both leave the standard persona-vector-on-context readout near zero or negative throughout so the gain is the *map*, not the readout family.
- **On synthetic prompts the context already gives the behavior away,** so the map buys nothing there it earns its keep exactly where the prompt does *not* telegraph the answer.

IMPLICATIONS & FUTURE WORK

- Much of an LLM's behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool, and evidence for the **persona selection model**.
- Theoretical analysis + dynamical systems.
- Predict other properties: correctness.
- Predicting effect of fine-tuning.
- Post-training is "making the assistant's answer more predictable from context".
- Automated redteaming / jailbreaking.

REFERENCES

[1] Chen et al. Persona vectors. 2025. [2] Betley et al. Emergent misalignment. 2025. [3] Marks et al. The persona selection model. 2026. [4] Lu et al. The assistant axis. 2026. [5] Hosseini & Fedorenko. Trajectory straightening. 2023. [6] Park et al. The linear representation hypothesis. 2024.

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